Guide to set up Pi Devices

1. Every Pi device in the network **should always run the registration script during first boot**, the file is test_device_registration_dual_sync.js located in git under rutag_backend/pi/device registration/.
2. After a successful run of the device registration script a file will be created device_id.txt. **This file must be retained. Each device uuid will be stored here, and this file is not to be copied to any other Pi device.** (Although this file can be copied within the pi to other directories to run other scripts).
3. A serviceAccountKey file which can be downloaded from firebase account of this project is required to run scripts to control sensor, send alerts and camera streaming. This file is not shared through git due to security concerns.
4. For camera control the directory in which your camera script runs should contain your serviceAccountKey file and a .env file. A typical .env file contents is shown in the image below

```
D: > .env
FIREBASE_KEY_PATH=./serviceAccountKey.json
FIREBASE_DB_URL=https://rutag-app-default-rtdb.asia-southeast1.firebasedatabase.app
DEVICE_ID=a8e2cbc7-1e25-47bb-ba63-0551e3e021a6
WEBRTC_WIDTH=640
WEBRTC_HEIGHT=480
WEBRTC_FPS=30
WEBRTC_BITRATE=1000k
WEBRTC_PREVIEW=true
```

   The bitrate , width, height, fps can be changed as per your preference and capacity of your Pi.
5. A typical .env file content of the sensor control script is shown below for your reference. This file should be in the directory alongwith serviceAccountKey file where your sensor control script is.

```
D: > dht > .env
SENSOR_CONTROL_API_URL=http://13.205.201.82/sensor-api
DEVICE_ID_FILE=./device_id.txt
API_KEY=admin_009db543d77b6639e42e947a6281fb5668cc92c4e6e89d241d318a0212549e38
DEVICE_REGISTRATION_SECRET=your-strong-secret-here
SENSOR_NAME=DHT11 Sensor
SENSOR_TYPE=dht11
SENSOR_STATUS_CHECK_INTERVAL_MS=5000
SENSOR_LOCAL_PORT=5000
DHT_GPIO_PIN=18
```

6. In case there is shutdown or restart of Pi , there is no requirement to register the device again.